GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005061

Subject Name: Polymer Technology

Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-04

Prerequisite:	Foundational knowledge of Polymer Technology, monomers, unit operations, and unit processes is required.
Rationale:	Polymers have become an un-detachable part of our everyday life. Could you imagine life in the world without polymer products? However, the awareness quotient tends to be on the lower side. This can be enhanced through specific courses designed for the purpose. The recent developments of Polymer Technology have revolutionized the field of material science increasing the use of polymer based substances from building materials to Packing materials, Fancy decoration articles, Electrical engineering, Communications, Automobile, Aircrafts, etc. Polymer Technology has carved a niche in the fields of electronics and electrical materials, textiles, aerospace industry, automobile industry, etc. The course aims to provide the students with profound educational activities. The Diploma chemical engineers have to deal with identification and characterization of raw material to ensure the quality of polymer product along with different techniques of processing. The content of this course is designed to enable diploma holders to develop the skills required for working in production, processing, testing, marketing and sales department of plastics, rubbers and fibre manufacturing Industries. These areas have been highlighted as they represent the needs of the industry.

Course Outcome:

After Completion of the Course, Student will able to:

No.	Course Outcomes	RBT Level
1	Know the polymer and its base, and select appropriate techniques of polymerization.	R,A
2	Produce plastics using appropriate reactions and unit operations steps.	R, A, U
3	Produce rubbers using appropriate reactions and unit operations steps	R, A, U

4	Produce fiber & resins using appropriate reactions and unit operations steps.	R, A, U
5	Apply selected different polymer processing techniques and Understand polymer degradation.	A

*Revised Bloom's Taxonomy (RBT)

Teaching and Examination Scheme:

Teaching Scheme(in Hours)			Total Credits L+T+(PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial/Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit	Topics and Sub-topics	No. of Hours	% of Weightage
Unit – I Fundamental of Polymer technology	**1.1 Definition and examples of monomer with their polymers.**	5	10
	1.2 Classification of polymers 1.2.1 Natural and synthetic polymers 1.2.2 Organic and in organic polymers 1.2.3 Thermoplastic and Thermosetting 1.2.4 Plastic, rubber, fibre and resin 1.2.5 Homo polymer and copolymer		
	1.3 Types of polymerization reaction 1.3.1 Addition polymerization 1.3.2 Condensation polymerization		
	1.4 Techniques of polymerization 1.4.1 Bulk polymerization, 1.4.2 Solution polymerization 1.4.3 Suspension polymerization, 1.4.4 Emulsion polymerization		

Unit – II Properties of Polymers	**2.1 General properties of Polymers** 2.1.1 Molecular weight 2.1.2 Crystallinity 2.1.3 Glass transition temperature **2.2 Specific Properties of Plastics** 2.2.1 Electrical Resistance 2.2.2 Chemical Resistance 2.2.3 Thermal Stability 2.2.4 Biodegradability **2.3 Specific Properties of Rubbers** 2.3.1 Elasticity 2.3.2 Electric Conductivity 2.3.3 Resistance to fatigue **2.4 Specific Properties of Fiber** 2.4.1 Resiliency 2.4.2 Elongation 2.4.3 Strength	5	7
Unit – III Plastics and manufacturing	**3.1 Properties , applications and manufacturing of different plastics:** 3.1.1 Polyethylene by Ziegler low pressure process for HDPE (High density polyethylene) 3.1.2 Polystyrene 3.1.3 Polytetrafluoro Ethylene (PTTE/teflon) 3.1.4 Melamine	5	14
Unit – IV Rubbers and manufacturing	**4.1 Properties, applications and manufacturing of different rubbers**: 4.1.1 Ethylene Propylene Terpolymers, 4.1.2 Butyl rubber 4.1.3 Polyurethane rubber 4.1.4 Silicone rubber	5	14
Unit – V Fibres and manufacturing	**5.1 Properties, applications and manufacturing of different fibres:** 5.1.1 Viscose rayon fibres 5.1.2 Cellulose Acetate fibres 5.1.3 Nylon-6	5	14

	5.1.4 Glass fibres		
Unit – VI Resins and manufacturing	**6.1 Classification of Resin** **6.2 Properties, applications and manufacturing of different resins:** 6.2.1 Phenol formaldehyde 6.2.2 Polyvinyl Resin 6.2.3 Urea formaldehyde	5	14
Unit – VI Polymer Processing	**7.1 introduction to polymer processing** **7.2 Various types of molding** 7.2.1 compression molding 7.2.2 injection molding 7.2.3 blow molding 7.2.4 extrusion **7.3 die casting** **7.4 calendaring** **7.5 fibre spinning** 7.5.1 wet spinning 7.5.2 dry spinning 7.5.3 melt spinning **7.6 vulcanization** **7.7 compounding** **7.8 coating** **7.9 mastication and mixing**	12	21
Unit – VII Polymer Degradation	**8.1 Introduction to degradation** **8.2 Types of degradation** 8.2.1 Thermal degradation 8.2.2 Ultrasonic degradation 8.2.3 Mechanical degradation.	3	4
TOTAL		45	100

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
30%	**35%**	**35%**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

Suggested Course Practical List:

Sr. No.	Practical Outcomes (PrOs)
1	Prepare classification chart based on properties and uses.
2	Demonstrate thermosetting and thermoplastic properties
3	Test the effects of solvents on plastic, rubber and fibre
4	Test the effects of acids and alkalies on plastic, rubber and fibre
5	Test effects of temperature on plastic, rubber and fibre
6	Determination of Density of Polymer Sample
7	Determine the acid value of glyptal resin
8	Prepare thermo-plastics (eg. PMMA, PE)
9	Prepare Polystyrene from styrene
10	Prepare Phenol Formaldehyde resin from Phenol
11	Prepare Urea Formaldehyde resin from Urea
12	Prepare Melamine Formaldehyde
13	Prepare cellulose acetate from cellulose
14	Perform Vulcanization of Rubber
15	Determination of Softening Point (Softening Temperature)
16	Prepare Glyptal resin

References / Suggested Learning Resources:

Books:

S. No.	Title of Books	Author	Publication	ISBN
1	Outlines of Chemical Technology (for the 21st Century)	Gopal M. Rao & Marshall Sittig (original by Dryden)	Affiliated East-West Press, New Delhi	ISBN-13: 9788123905730
2	A Textbook of Chemical Technology, Vol. I & II	G.N. Pandey	Vikas Publishing House	Vol. I: ISBN-13: 9780706986280; Vol. II: ISBN-13: 9780706986297
3	Shreve's Chemical Process Industries (5th Edition)	George T. Austin (based on D.D. Shreve's original work)	McGraw-Hill Education	ISBN-13: 9781259029455
4	Polymer science	V.R. Gowariker, N.V. Viswanathan, Jayadev Sreedhar	New Age International Pvt. Ltd., New Delhi	ISBN-13: 9788122436907
5	Polymer science and Technology	Joel R.; Fried	PHI Learning, New Delhi, 2007, 2nd Edition	ISBN-13: 9780137039555
6	Text on petrochemicals	B.K. Bhaskararao	Khanna Publishers, Delhi, 1998, 2nd Edition	ISBN-13: 9788174090447
7	A Hand Book of Polymer Science (For B.Tech., B.Sc. and Diploma Courses of Indian Universities)	Dr. S. Khalid Hasan & Fouzia Usmani	Paperback, 2020	ISBN-13: 9789388965282
8	The Complete Technology Book on Industrial Polymers, Additives, Colourants and Fillers	NIIR Board of Consultants & Engineers	Asia Pacific Business Press Inc. (2006)	ISBN-10: 8178330203 / ISBN-13: 9788178330206

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